

Trạng thái	Đã xong
Bắt đầu vào lúc	Thứ Sáu, 23 tháng 8 2024, 1:55 PM
Kết thúc lúc	Thứ Sáu, 23 tháng 8 2024, 1:57 PM
Thời gian thực hiện	1 phút 33 giây
Điểm	5,00/5,00
Điểm	10,00 trên 10,00 (100%)



Câu hỏi 1

Đúng

Đạt điểm 1,00 trên 1,00

In the coordinate plane, we have class Point to store a point with it's x-y coordinate.

Your task in this exercise is to implement functions marked with `/* * STUDENT ANSWER */`.

Note: For exercises in Week 1, we have `#include <bits/stdc++.h>` and using namespace std;

For example:

Test	Result
Point A(2, 3); cout << A.getX() << " " << A.getY();	2 3
Point A(2, 3); Point B(1, 1); cout << pow(A.distanceToPoint(B), 2);	5

Answer: (penalty regime: 0 %)

Reset answer

```

1 class Point
2 {
3     private:
4         double x, y;
5
6     public:
7         Point()
8         {
9             /*
10              * STUDENT ANSWER
11              * TODO: set zero x-y coordinate
12              */
13             x=y=0;
14         }
15
16         Point(double x, double y)
17         {
18             /*
19              * STUDENT ANSWER
20              */
21             this->x=x;
22             this->y=y;
23         }
24
25         void setX(double x)
26         {
27             /*
28              * STUDENT ANSWER
29              */
30             this->x =x;
31         }
32
33         void setY(double y)
34         {
35             /*
36              * STUDENT ANSWER
37              */
38             this->y=y;
39         }
40
41         double getX() const
42         {
43             /*
44              * STUDENT ANSWER

```

```

45         */
46         return this->x;
47     }
48
49     double getY() const
50     {
51         /*
52          * STUDENT ANSWER
53          */
54         return this->y;
55     }
56
57     double distanceToPoint(const Point& pointA)
58     {
59         /*
60          * STUDENT ANSWER
61          * TODO: calculate the distance from this point to point A in the coordinate plane
62          */
63         int x = pointA.x;
64         int y = pointA.y;
65         double distance = sqrt(pow((x-this->x),2) + pow((y-this->y),2));
66         return distance;
67     }
68 }
69 };

```

	Test	Expected	Got	
✓	Point A(2, 3); cout << A.getX() << " " << A.getY();	2 3	2 3	✓
✓	Point A(2, 3); Point B(1, 1); cout << pow(A.distanceToPoint(B), 2);	5	5	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.



Câu hỏi 2

Đúng

Đạt điểm 1,00 trên 1,00

In the coordinate plane, a circle is defined by center and radius.

Your task in this exercise is to implement functions marked with `/* * STUDENT ANSWER */`.

Note: you can use implemented class Point in *previous question*

For example:

Test	Result
Circle A; A.printCircle();	Center: {0.00, 0.00} and Radius 0.00

Answer: (penalty regime: 0 %)

Reset answer

```

1  #include<iostream>
2  #include<string>
3  #include<cmath>
4
5
6  class Point
7  {
8  private:
9      double x, y;
10
11 public:
12     Point()
13     {
14         /*
15          * STUDENT ANSWER
16          * TODO: set zero x-y coordinate
17          */
18         x=y=0;
19     }
20
21     Point(double x, double y)
22     {
23         /*
24          * STUDENT ANSWER
25          */
26         this->x=x;
27         this->y=y;
28     }
29
30     void setX(double x)
31     {
32         /*
33          * STUDENT ANSWER
34          */
35         this->x =x;
36     }
37
38     void setY(double y)
39     {
40         /*
41          * STUDENT ANSWER
42          */
43         this->y=y;
44     }
45

```

```
46 double getX() const
47 {
48     /*
49     * STUDENT ANSWER
50     */
51     return this->x;
52 }
53
54 double getY() const
55 {
56     /*
57     * STUDENT ANSWER
58     */
59     return this->y;
60 }
61
62 double distanceToPoint(const Point& pointA)
63 {
64     /*
65     * STUDENT ANSWER
66     * TODO: calculate the distance from this point to point A in the coordinate plane
67     */
68     int x = pointA.x;
69     int y = pointA.y;
70     double distance = sqrt(pow((x-this->x),2) + pow((y-this->y),2));
71     return distance;
72 }
73 };
74
75
76 class Circle
77 {
78 private:
79     Point center;
80     double radius;
81
82 public:
83     Circle()
84     {
85         /*
86         * STUDENT ANSWER
87         * TODO: set zero center's x-y and radius
88         */
89         center.setX(0);
90         center.setY(0);
91         radius=0;
92     }
93
94     Circle(Point c, double radius)
95     {
96         /*
97         * STUDENT ANSWER
98         */
99         double x=c.getX();
100        double y=c.getY();
101
102        center.setX(x);
103        center.setY(y);
104
105        this->radius=radius;
106    }
107
108     Circle(const Circle &circle)
109     {
110         /*
111         * STUDENT ANSWER
112         */
113
114         double x=circle.center.getX();
115         double y=circle.center.getY();
116
117         double radius= circle.getRadius();
118     }
```

```

118
119     this->center.setX(x);
120     this->center.setY(y);
121
122     this->radius=radius;
123
124 }
125
126 void setCenter(Point point)
127 {
128
129     double x=point.getX();
130     double y=point.getY();
131
132     this->center.setX(x);
133     this->center.setY(y);
134 }
135
136 void setRadius(double radius)
137 {
138     /*
139     * STUDENT ANSWER
140     */
141     this->radius=radius;
142 }
143
144 Point getCenter() const
145 {
146     /*
147     * STUDENT ANSWER
148     */
149     return this->center;
150 }
151
152 double getRadius() const
153 {
154     /*
155     * STUDENT ANSWER
156     */
157     return radius;
158 }
159
160 void printCircle()
161 {
162     printf("Center: {%.2f, %.2f} and Radius %.2f\n", this->center.getX(), this->center.getY(), this->radius);
163 }
164 };

```

	Test	Expected	Got	
✓	Circle A; A.printCircle();	Center: {0.00, 0.00} and Radius 0.00	Center: {0.00, 0.00} and Radius 0.00	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 3

Đúng

Đạt điểm 1,00 trên 1,00

In a game, we have class **Character** to store characters' data.
The class **Character** is declared as below:

```
class Character {
protected:
    int hp;
    int x;
    int y;
public:
    // Constructor: set the values of x and y and hp to 0
    Character();

    // Constructor: set the values of hp, x and y to each parameter
    Character(int hp, int x, int y);

    // Set and get hp
    int getHp();
    void setHp(int hp);

    // Set and get x
    int getX();
    void setX(int x);

    // Set and get y
    int getY();
    void setY(int y);

    // Get Manhattan distance to other character
    int getManhattanDistTo(Character* other);
};
```

Your task is to define the constructors and the methods of the class.

Note:

In this task, **iostream** library has been included, and **namespace std** is being used. No other libraries are allowed.

For example:

Test	Result
Character ch1(100, 3, 6); cout << ch1.getHp() << " " << ch1.getX() << " " << ch1.getY();	100 3 6

Answer: (penalty regime: 0 %)

[Reset answer](#)

```
1 #include<cmath>
2
3 Character::Character() {
4     // STUDENT ANSWER
5     x=y=hp=0;
6 }
7
8 Character::Character(int hp, int x, int y) {
```

```

8 Character::Character(int hp, int x, int y) {
9     // STUDENT ANSWER
10    this->hp=hp;
11    this->x=x;
12    this->y=y;
13 }
14
15 int Character::getHp() {
16     // STUDENT ANSWER
17     return hp;
18 }
19
20 void Character::setHp(int hp) {
21     // STUDENT ANSWER
22     this->hp=hp;
23 }
24
25 int Character::getX() {
26     // STUDENT ANSWER
27     return this->x;
28 }
29
30 void Character::setX(int x) {
31     // STUDENT ANSWER
32     this->x=x;
33 }
34
35 int Character::getY() {
36     // STUDENT ANSWER
37     return this->y;
38 }
39
40 void Character::setY(int y) {
41     // STUDENT ANSWER
42     this->y=y;
43 }
44
45 int Character::getManhattanDistTo(Character* other) {
46     // STUDENT ANSWER
47     int a,b;
48     a= other->getX();
49     b=other->getY();
50
51     int distance= abs(a-this->x) + abs(b-this->y);
52     return distance;
53 }

```

	Test	Expected	Got	
✓	Character ch1(100, 3, 6); cout << ch1.getHp() << " " << ch1.getX() << " " << ch1.getY();	100 3 6	100 3 6	✓
✓	Character ch2; cout << ch2.getHp() << " " << ch2.getX() << " " << ch2.getY();	0 0 0	0 0 0	✓
✓	Character* ch31 = new Character(100, 1, 2); Character* ch32 = new Character(100, -3, 4); cout << ch31->getManhattanDistTo(ch32); delete ch31; delete ch32;	6	6	✓
✓	Character ch4; ch4.setX(4); cout << ch4.getX();	4	4	✓

	Test	Expected	Got	
✓	Character ch5; ch5.setY(5); cout << ch5.getY();	5	5	✓
✓	Character ch6; ch6.setHp(6); cout << ch6.getHp();	6	6	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.



Câu hỏi 4

Đúng

Đạt điểm 1,00 trên 1,00

Hoang is a K19 student studying at Bach Khoa University. He plans to write a book management software for the library. In the class design, Hoang has designed the class Book as follows:

```
class Book
{
private:
    char* title;
    char* authors;
    int publishingYear;
public:
    // some method
}
```

Your task in this exercise is to implement functions marked with `/* * STUDENT ANSWER */`.

Note: For exercises in Week 2, we have `#include <bits/stdc++.h>` and using namespace std;

For example:

Test	Result
Book book1("Giai tich 1","Nguyen Dinh Huy",2000); book1.printBook();	Giai tich 1 Nguyen Dinh Huy 2000
Book book1("Giai tich 1","Nguyen Dinh Huy",2000); Book book2 = book1; book2.printBook();	Giai tich 1 Nguyen Dinh Huy 2000

Answer: (penalty regime: 0 %)

Reset answer

```
1 #include<iostream>
2 #include<cstring>
3
4 using namespace std;
5
6
7
8 class Book
9 {
10 private:
11     char* title;
12     char* authors;
13     int publishingYear;
14
15 public:
16     Book()
17     {
18         /*
19          * STUDENT ANSWER
20          * TODO: set zero publishingYear and null pointer
21          */
22         title=authors=nullptr;
23         publishingYear=0;
24     }
25
26     Book(const char* title, const char* authors, int publishingYear)
```

```

20     book(const char* title, const char* authors, int publishingYear)
21     {
22     /*
23      * STUDENT ANSWER
24      */
25
26
27
28
29
30
31
32
33
34         this->title= strdup(title);
35         this->authors= strdup(authors);
36         this->publishingYear=publishingYear;
37
38     }
39
40     Book(const Book &book)
41     {
42     /*
43      * STUDENT ANSWER
44      * TODO: deep copy constructor
45      */
46         char*a= book.getTitle();
47         char*b= book.getAuthors();
48         int c = book.getPublishingYear();
49
50         this-> title =a;
51         this->authors=b;
52         this->publishingYear=c;
53
54     }
55
56     void setTitle(const char* title)
57     {
58
59         this->title= strdup(title);
60
61
62     }
63
64     void setAuthors(const char* authors)
65     {
66     /*
67      * STUDENT ANSWER
68      */
69     this->authors= strdup(authors);    }
70
71     void setPublishingYear(int publishingYear)
72     {
73     /*
74      * STUDENT ANSWER
75      */
76         this->publishingYear=publishingYear;
77
78     }
79
80     char* getTitle() const
81     {
82     /*
83      * STUDENT ANSWER
84      */
85         return this->title;
86     }
87
88     char* getAuthors() const
89     {
90     /*
91      * STUDENT ANSWER
92      */
93         return this->authors;
94
95     }
96
97     int getPublishingYear() const
98     {
99     /*

```

```

99
100     * STUDENT ANSWER
101     */
102     return this->publishingYear;
103 }
104
105 ~Book()
106 {
107     /*
108     * STUDENT ANSWER
109     */
110     title=authors=nullptr;
111     publishingYear=0;
112 }
113
114 void printBook(){
115     printf("%s\n%s\n%d", this->title, this->authors, this->publishingYear);
116 }
117 };

```

	Test	Expected	Got	
✓	Book book1("Giai tich 1","Nguyen Dinh Huy",2000); book1.printBook();	Giai tich 1 Nguyen Dinh Huy 2000	Giai tich 1 Nguyen Dinh Huy 2000	✓
✓	Book book1("Giai tich 1","Nguyen Dinh Huy",2000); Book book2 = book1; book2.printBook();	Giai tich 1 Nguyen Dinh Huy 2000	Giai tich 1 Nguyen Dinh Huy 2000	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.



Câu hỏi 5

Đúng

Đạt điểm 1,00 trên 1,00

1. In the toy store, all toy has a price. Car toy has a price and color, Puzzle toy has a price and size. We have to implement class CarToy and class PuzzleToy which inherit from class Toy.
2. class ToyBox has a pointer array to store a list of toys (up to 5 items including car and puzzle) and number of items in the box.

Your task is to implement two function addItem(...) in class ToyBox. If successfully added, the function returns the current number of toys in the box. If the box is full, return -1.

For example:

Test	Result
CarToy car(20000,red); PuzzleToy puzzle(30000,small); car.printType(); puzzle.printType();	This is a car toy This is a puzzle toy
CarToy car(20000,red); PuzzleToy puzzle(30000,small); ToyBox box; box.addItem(car); box.addItem(puzzle); box.printBox();	This is a car toy This is a puzzle toy
Toy* toy = new CarToy(30000,red); toy->printType();	This is a car toy

Answer: (penalty regime: 0 %)

Reset answer

```

1  #include<iostream>
2  using namespace std;
3
4  enum Color
5  {
6      red,
7      green,
8      blue
9  };
10 enum Size
11 {
12     small,
13     medium,
14     big
15 };
16
17 class Toy
18 {
19     protected:
20         double price;
21
22     public:
23         Toy(double price)
24         {
25             this->price = price;
26         }

```

```
26     },
27
28     virtual void printType() = 0;
29     friend class ToyBox;
30 };
31
32 class CarToy : public Toy
33 {
34 private:
35     Color color;
36
37 public:
38     CarToy(double price, Color color) : Toy(price)
39     {
40         /*
41          * STUDENT ANSWER
42          */
43         this->price=price;
44         this->color=color;
45     }
46
47     void printType()
48     {
49         cout << "This is a car toy\n";
50     }
51
52     friend class ToyBox;
53 };
54
55 class PuzzleToy : public Toy
56 {
57 private:
58     Size size;
59
60 public:
61     PuzzleToy(double price, Size size) : Toy(price)
62     {
63         /*
64          * STUDENT ANSWER
65          */
66         this->price=price;
67         this->size=size;
68     }
69
70     void printType()
71     {
72         cout << "This is a puzzle toy\n";
73     }
74
75     friend class ToyBox;
76 };
77
78 class ToyBox
79 {
80 private:
81     Toy* toyBox[5];
82     int numberOfItems;
83
84 public:
85     ToyBox()
86     {
87         /*
88          * STUDENT ANSWER
89          * TODO: set zero numberOfItems and nullptr toyBox
90          */
91
92         toyBox[0]=nullptr;
93
94         numberOfItems =0;
95     }
96
97     int addItem(const CarToy& carToy)
98     {
```

```

100  /*
101  * STUDENT ANSWER
102  * TODO: function add a new Car toy to the box.
103  *       If successfully added, the function returns the current number of toys in the box.
104  *       If the box is full, return -1.
105  */
106
107  if (numberOfItems==5) return -1;
108  else {
109      toyBox[numberOfItems]= new CarToy(carToy);
110      numberOfItems++;
111      return numberOfItems;
112  }
113
114
115  }
116
117  int addItem(const PuzzleToy& puzzleToy)
118  {
119      /*
120      * STUDENT ANSWER
121      * TODO: function add a new Puzzle toy to the box.
122      *       If successfully added, the function returns the current number of toys in the box.
123      *       If the box is full, return -1.
124      */
125
126      if (numberOfItems==5) return -1;
127      else{
128          toyBox[numberOfItems]= new PuzzleToy(puzzleToy);
129          numberOfItems++;
130          return numberOfItems;
131      }
132
133  }
134
135
136  void printBox()
137  {
138      for (int i = 0; i < numberOfItems; i++)
139          toyBox[i]->printType();
140  }
141  };

```

	Test	Expected	Got	
✓	CarToy car(20000,red); PuzzleToy puzzle(30000,small); car.printType(); puzzle.printType();	This is a car toy This is a puzzle toy	This is a car toy This is a puzzle toy	✓
✓	CarToy car(20000,red); PuzzleToy puzzle(30000,small); ToyBox box; box.addItem(car); box.addItem(puzzle); box.printBox();	This is a car toy This is a puzzle toy	This is a car toy This is a puzzle toy	✓
✓	Toy* toy = new CarToy(30000,red); toy->printType();	This is a car toy	This is a car toy	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.

